

Physics-Guided Reduced-Order Machine Learning for Fast Structural Deformation Prediction

Course: Computational Mechanics

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1. Problem Statement and Objectives

- The Finite Element Method (FEM) is commonly used to study how structures deform under different loads and material conditions.
- FEM provides reliable results, but detailed models can contain a large number of unknowns.
- Solving the same model repeatedly for different conditions can therefore require considerable computational time.
- This project explores a faster approach by combining FEM, model reduction, and machine learning.
- A cantilever beam is used as the initial example because its geometry and boundary conditions are simple to analyse.
- FEM simulations will be performed for different values of:
 - Applied force
 - Young's modulus
 - Beam thickness
 - Other relevant geometric parameters
- The resulting displacement fields will be analysed using Proper Orthogonal Decomposition (POD) and Singular Value Decomposition (SVD).
- Instead of predicting every nodal displacement directly, the machine-learning model will predict a small number of reduced-order coefficients.
- These coefficients will then be combined with the dominant deformation modes to reconstruct the complete displacement field.

Main Objectives

- Develop a FEM model for a cantilever beam.
- Generate displacement data for different physical and geometric conditions.
- Organise the displacement results into a snapshot matrix.
- Use SVD/POD to identify the most important deformation modes.
- Select a suitable number of modes without losing significant information.
- Calculate the reduced-order coefficients.
- Train a regression model to predict these coefficients.
- Reconstruct the displacement field from the predicted coefficients.
- Compare the reconstructed result with the full FEM solution.
- Study the balance between prediction accuracy and computational efficiency.

2. Literature Review

2.1 POD and SVD-Based Model Reduction

- Proper Orthogonal Decomposition is used to describe complex physical behaviour using a smaller number of important patterns.
- In this project, each FEM simulation produces a displacement vector.
- These vectors are arranged as columns of a snapshot matrix:

$$X = [u_1 \ u_2 \ \dots \ u_m]$$

- SVD decomposes the snapshot matrix as:

$$X = U \Sigma V^T$$

- The columns of U represent the main deformation modes.
- The singular values in Σ indicate how important those modes are.
- A larger singular value means that the corresponding mode contributes more to the displacement data.
- The energy captured by the first k modes is calculated using:

$$E_k = [\Sigma_{i=1}^k \sigma_i^2] / [\Sigma_{i=1}^r \sigma_i^2]$$

- The first k modes are selected to form the reduced basis:

$$\Phi = [\phi_1, \phi_2, \dots, \phi_k]$$

- The displacement field can then be approximated as:

$$u \approx \Phi a$$

- Here, a is a small vector containing the contribution of each selected mode.
- The main advantage is that a displacement field with many nodal values can be represented using only a few coefficients.
- The number of modes must be chosen carefully:
 - Too few modes may reduce accuracy.
 - Too many modes may make the machine-learning model unnecessarily complex.

2.2 Reduced-Order Modelling

- Reduced-order modelling is useful when a detailed numerical model must be solved many times.
- Instead of repeating the complete FEM calculation, the reduced model attempts to preserve the most important structural behaviour using fewer variables.
- The POD modes in this project are obtained directly from FEM displacement data.
- The quality of the reduced model depends strongly on the range and quality of the FEM simulations.
- If a new input lies far outside the training range, the reduced model may not predict the deformation accurately.

2.3 Machine Learning and Physics Guidance

- Machine learning can learn how physical parameters affect structural deformation.
- Directly predicting every nodal displacement may result in a large and difficult output.
- This project instead predicts the reduced coefficients:

$$[F, E, t, \dots] \rightarrow \text{ML model} \rightarrow [a_1, a_2, \dots, a_k]$$

- The complete displacement field is reconstructed using:

$$\hat{u} = \Phi \hat{a}$$

- The approach is physics-guided because:
 - The training data are produced using FEM.
 - The POD modes are extracted from physically meaningful displacement fields.
 - The machine-learning model predicts coefficients associated with these FEM-derived modes.
- The governing equations are not directly included in the machine-learning loss.
- A fully physics-informed neural network could be considered as an extension of this work.

3. Proposed Methodology

- The project follows the workflow:

FEM → Displacement Snapshots → SVD/POD → Reduced Coefficients → Machine Learning →
Displacement Reconstruction

3.1 Cantilever Beam Model

- A cantilever beam is selected as the first test problem.

- One end of the beam is fixed.
- A load is applied close to the free end.
- The main parameters considered are:
 - Applied force, F
 - Young's modulus, E
 - Beam thickness, t
 - Other geometric parameters, if needed
- These parameters influence the stiffness and deformation of the beam.
- The simple setup makes it easier to study the complete model-reduction process.

3.2 FEM Formulation

- The beam is divided into a number of finite elements.
- Each element contributes to the global stiffness matrix.
- The structural equilibrium equation is:

$$Ku = f$$

- (K) represents the global stiffness matrix.
- (u) represents the nodal displacement vector.
- (f) represents the applied force vector.
- The fixed boundary conditions are applied at the supported end.
- The equation is solved for several combinations of input parameters.
- Each solution provides one displacement snapshot for the reduced-order model.

3.3 Generation of Displacement Snapshots

- The FEM model is solved repeatedly for different input conditions.
- Each simulation produces a displacement vector u_j .
- These displacement vectors are collected in the snapshot matrix:

$$X = [u_1 \ u_2 \ \dots \ u_m]$$

- All simulations must use:
 - The same mesh
 - The same degree-of-freedom ordering
 - The same displacement format
- This ensures that corresponding entries represent the same physical location in every simulation.
- The available cases are divided into:
 - Training data
 - Testing data
- The testing cases are kept separate so that the model can be evaluated on conditions it has not seen during training.

3.4 SVD and POD Mode Selection

- SVD is applied to the snapshot matrix:

$$X = U\Sigma V^T$$

- The leading columns of U are taken as the POD modes.
- The number of modes is selected by examining:
 - Cumulative energy
 - Reconstruction error
- Retaining more modes usually improves the representation of the displacement field.
- However, it also increases the number of coefficients that the machine-learning model must predict.
- The final mode count should therefore provide a practical balance between accuracy and simplicity.

3.5 Calculation of Reduced Coefficients

- The displacement field is represented using the selected POD modes:

$$u \approx \Phi a$$

- Since the POD modes are orthonormal, the coefficients are calculated as:

$$a = \Phi^T u$$

- Each coefficient describes how strongly one POD mode contributes to the displacement.
- These coefficients are used as the target outputs for the machine-learning model.
- This reduces the prediction problem from many nodal displacements to only a few meaningful values.

3.6 Machine-Learning Model

- A regression model is used because the reduced coefficients are continuous numerical quantities.
- The inputs are the physical and geometric parameters.
- The outputs are the reduced coefficients:

$$[F, E, t, \dots] \rightarrow \text{ML model} \rightarrow [a_1, a_2, \dots, a_k]$$

- The model is trained using the FEM-generated input-output pairs.
- Its performance is tested using parameter combinations that were not included during training.
- The final regression method will depend on the amount of available data and the required accuracy.

3.7 Reconstruction and Validation

- For a new set of input parameters, the trained model predicts the reduced coefficients $\hat{a}$.
- The displacement field is then reconstructed using:

$$\hat{u} = \Phi \hat{a}$$

- The reconstructed field is compared with the corresponding FEM solution.
- The relative error is calculated as:

$$e_{\text{rel}} = \|u - \hat{u}\|_2 / \|u\|_2$$

- The root mean square error is calculated as:

$$\text{RMSE} = \sqrt{\{(1/n) \sum_{j=1}^n (u_j - \hat{u}_j)^2\}}$$

- The evaluation will consider:
 - Relative displacement error
 - RMSE
 - Maximum displacement error
 - Similarity between FEM and reconstructed deformation
 - Prediction time
 - Performance on unseen input conditions

3.8 Comparison with Full FEM

- The main comparison will be between:
 - The full FEM solution
 - The reduced-order machine-learning prediction
- The comparison will focus on:
 - Accuracy
 - Reconstruction error
 - Computational time
 - Number of retained modes
 - Model complexity
 - Generalisation to new conditions
- Generating FEM snapshots, calculating POD modes, and training the model form the offline stage.

- Predicting new coefficients and reconstructing the displacement from the online stage.
- The main purpose of the reduced model is to make repeated predictions faster after the offline stage is completed.

4. Individual Contribution

- My main contribution is the mathematical model-reduction stage of the project.
- I will study how the high-dimensional displacement data from FEM can be represented using a smaller number of meaningful modes.
- My responsibilities include:
 - Understanding the FEM degrees of freedom and displacement vectors.
 - Constructing the displacement snapshot matrix.
 - Applying SVD to the snapshot data.
 - Interpreting U , Σ , and V^T .
 - Studying the singular values and their importance.
 - Selecting the number of POD modes.
 - Calculating the reduced coefficients.
 - Measuring reconstruction error for different mode counts.
 - Providing the coefficient data required for machine-learning training.
- My work forms the link between the FEM simulation stage and the machine-learning stage.
- The main equations used in my work are:

$$X = U\Sigma V^T$$

$$E_k = [\sum_{i=1}^k \sigma_i^2] / [\sum_{i=1}^r \sigma_i^2]$$

$$a = \Phi^T u$$

$$u \approx \Phi a$$

5. Anticipated Results

- The first few POD modes are expected to capture most of the important deformation behaviour of the beam.
- The displacement field should be represented using far fewer variables than the original FEM model.
- The machine-learning model is expected to predict the reduced coefficients accurately within the selected training range.
- The reconstructed displacement should be close to the FEM solution for unseen cases within that range.
- The reduced-order model is expected to produce predictions faster than repeatedly solving the full FEM model.
- The final performance will depend on:
 - Number of FEM simulations
 - Quality of the training data
 - Range of physical parameters
 - Number of retained POD modes
 - Selected machine-learning method
- Predictions may become less reliable outside the parameter range used for training.
- Both accuracy and generalisation will therefore be tested.

6. Conclusion and Future Work

- This project combines FEM, SVD/POD, and machine learning to develop a faster method for predicting structural deformation.
- FEM provides the displacement data needed to understand the beam's response.
- SVD/POD identifies the most important deformation patterns.
- Machine learning predicts the reduced coefficients associated with these patterns.
- The predicted coefficients are used to reconstruct the complete displacement field.

- The reduced model will be assessed against the full FEM solution.
- The main performance measures are:
 - Prediction accuracy
 - Reconstruction error
 - Computational time
 - Performance on unseen cases
- Future work may include:
 - Testing more complex geometries.
 - Studying nonlinear structural behaviour.
 - Including more loading conditions.
 - Adding governing equations or physical residuals to the learning process.
 - Applying the method to real-time structural monitoring.
 - Exploring digital-twin applications.
 - Quantifying uncertainty in material properties, loads, and measurements.

7. References

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